Mechatronic Sensors Trainer

Fundamental Labs – Microphone & Speaker

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Mechatronic Sensors Trainer – Application Guide

Microphone & Speaker Lab

Why Explore Microphones and Speakers?

Microphones and speakers are fundamental devices for converting between sound waves and electrical signals. Understanding how they work provides insight into signal capture, processing, and playback, which are essential in technologies like voice assistants, communication systems, and audio interfaces. Exploring them helps engineers connect physical acoustics with digital signal processing and real-world sensing applications.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they output, when they are used, and be comfortable with using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**024_microphone_speaker**

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of **Microphones and Speakers**. The lab will have you explore the raw data from the microphone and how to write sound buffers to the speaker. Finally, you will perform meaningful analysis on the sound data in both time and frequency domain.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.